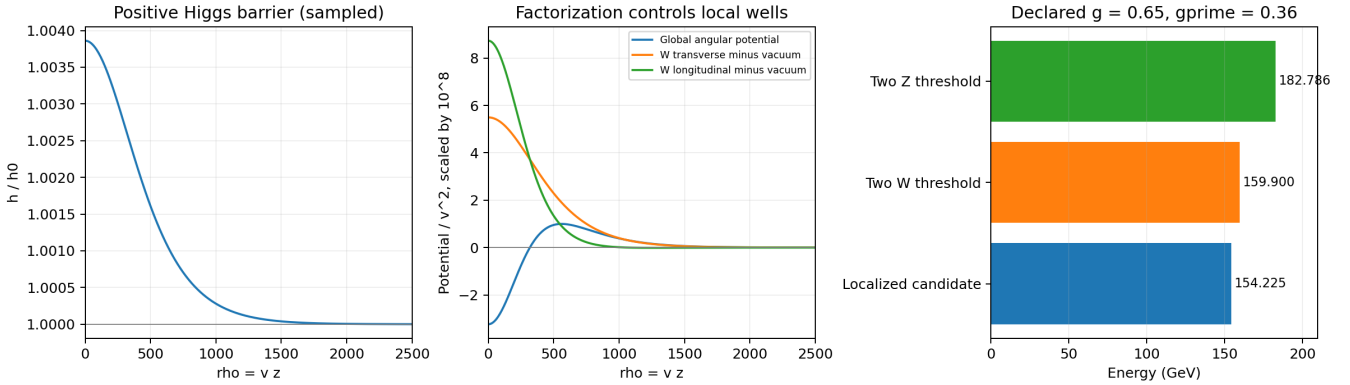


Goldstone and gauge channels of the localized wall candidate

This continuation takes item 19 of the requested expansion: the Goldstone and gauge channels of the radial wall mode. The complete Higgs-doublet and kinetic expansion has zero quadratic mixing from the source/radial sector into either angular Higgs fields or gauge fields. Goldstone-longitudinal derivative mixing is nevertheless present inside the separate gauge block. Exact first-order operator factorizations establish global angular positivity and a conditional vacuum-mass lower bound for both physical transverse and longitudinal vector modes. At the declared gauge inputs, the 154.225 GeV localized candidate is below the two-W and two-Z thresholds. Nonzero cubic scalar and vector vertices remain, and an integrated Ward identity checks them independently. The calculation therefore closes the quadratic leakage question for the supplied action while making the remaining nonlinear lifetime question explicit.



The full doublet expansion. Replace the real radial Higgs coordinate h by four canonical real fields: radial displacement η and three angular coordinates π_a . In the dimensionless square-completed potential, its radius squared is $(h+\eta)^2 + \sum \pi_a^2$. Differentiating before imposing $\pi_a = \eta = 0$ gives zero source-angular, singlet-angular and radial-angular Hessian entries. The three equal angular diagonal entries are

$$V_G = \lambda_H(h^2 - h_0^2) + \kappa(u^2 - 1) = \frac{h''}{h}.$$

Here primes mean ρ derivatives and the last equality uses the exact static radial background equation. The ungauged angular operator is consequently

$$L_G = -\partial_\rho^2 + \frac{h''}{h} = (-\partial_\rho - h'/h)(\partial_\rho - h'/h).$$

For a smooth positive h , this is a nonnegative quadratic form on compactly supported functions, extending to its natural self-adjoint form domain. A negative angular potential at a point does not imply an angular instability. The function h itself solves $L_G h = 0$. Since h approaches the nonzero vacuum h_0 , that zero-energy solution is a threshold resonance rather than an additional normalizable wall particle. In the ungauged $O(4)$ model the three angular channels are massless continua. After electroweak gauging they supply longitudinal vector degrees of freedom and must not be counted as three additional physical massless decay products.

What the kinetic expansion does and does not separate. For a single broken Abelian component, the relevant kinetic density can be written as one half of $[(\partial_\mu h + \partial_\mu \eta + q A_\mu \pi)^2 + (\partial_\mu \pi - q A_\mu (h + \eta))^2]$. Its quadratic mixed term is $q A_\mu$ times $[\pi \partial_\mu h - h \partial_\mu \pi]$. There is no radial-vector or radial-derivative-vector quadratic term. A varying Higgs background nevertheless retains Goldstone-vector derivative mixing. Gauge fixing can reorganize that block but cannot justify discarding it simply because the radial mode does not couple to it linearly. The new symbolic checks retain this term explicitly. The electroweak neutral mass matrix has a constant massless photon direction; its radial mass vertex also vanishes exactly. Gauge self-interactions and charged-vector loops lie beyond the quadratic expansion.

Physical vector operators. When h is everywhere positive, a unitary-gauge Proca description is valid locally and globally on this background. Each broken gauge direction has mass $m(\rho)=c \ h(\rho)$, with $c=g/2$ for W and $c=\sqrt{(g^2+g'^2)}/2$ for Z in the dimensionless variables. Fourier transformation parallel to the wall leaves two physical polarizations governed by $L_T=-\partial_\rho^2+m^2$. The remaining physical polarization follows from the Proca constraint. Setting $F=m^2 A_\rho$, the constraint and field equation give $-F''+2(m'/m)F'+m^2 F=E F$. The canonical variable is $\chi=F/m=m A_\rho$. Its operator is

$$L_L = -\partial_\rho^2 + m^2 + 2(m'/m)^2 - m''/m.$$

The transformation is checked exactly by symbolic substitution, not by treating the longitudinal field as another minimally coupled scalar. With $a=m'/m$, the operator becomes

$$L_L = (-\partial_\rho + a)(\partial_\rho + a) + m^2.$$

This factorization retains the inhomogeneous longitudinal response. Its derivative term can make the local potential lower than the vacuum value even when h never falls below h_0 . The quadratic form remains the sum of a square and the positive mass term. The test suite includes precisely such a negative local well and independently integrates both representations of its quadratic form.

A conditional spectral theorem. If h is smooth, positive and at least h_0 everywhere, then m^2 is at least its vacuum value m_0^2 . For every compactly supported trial function, the transverse quadratic form is integral $[\text{absolute } f'^2 + m^2 \text{ absolute } f^2]$, and the longitudinal form is integral $[\text{absolute } (f' + a f)^2 + m^2 \text{ absolute } f^2]$. Both are at least m_0^2 times the norm. Thus neither physical vector operator supports a localized state below the vacuum threshold. This statement is exact conditional mathematics. The finite numerical profile supports the hypotheses, but its sampled signs do not constitute an interval-arithmetic proof of them.

There is a useful background maximum principle. Suppose a positive static h tends to h_0 and the source satisfies absolute u at most one. At an interior minimum with $h < h_0$, the static equation $h'' = h[\lambda_H(h^2 - h_0^2) + \kappa(u^2 - 1)]$ has a strictly negative right-hand side for positive λ_H and nonnegative portal. That contradicts the nonnegative second derivative at a minimum. Hence such a profile has h at least h_0 . Positivity and source boundedness remain explicit hypotheses; this argument does not certify the continuum signs of an arbitrary computed profile. Item 18, interval proofs of the profile signs, is therefore still a separate task.

The physical-scale candidate. Keep the preceding spectator inputs $v=30000$ GeV, heavy $M=300000$ GeV, Gaussian current $g_{\text{source}}=3000$ GeV, radial Higgs vacuum 246 GeV, Higgs quartic 0.13 and portal ten to the minus seven. The source quartic is retuned as in the localized-candidate calculation. Use declared gauge couplings $g=0.65$ and $g'=0.36$, which are benchmark inputs rather than a matched extraction at a specified experimental scale. The resulting vacuum W mass is 79.95 GeV and Z mass approximately 91.3932213 GeV. The two-particle thresholds are therefore 159.9 GeV and approximately 182.7864426 GeV. The reconstructed radial candidate mass is approximately 154.2250242 GeV. It is below both thresholds. Given the continuum profile hypotheses above, localized vector states cannot lower those thresholds either. No additional linear radial leakage occurs even when a single-vector continuum is energetically accessible, because the corresponding quadratic mixing vanishes.

Cubic vertices prevent a stronger stability conclusion. The exact scalar derivatives are $V_u \pi_a \pi_a = 2\kappa u$ and $V_h \pi_a \pi_a = 2\lambda_H h$. A radial/source fluctuation ψ therefore induces the dimensionless local global-angular pair vertex

$$c_{GG}(\rho) = 2\kappa u \psi_u + 2\lambda_H h \psi_h.$$

The vector mass Hessian has derivative $g^2 h/2$ for W and $(g^2+g'^2)h/2$ for Z . Multiplying these by ψ_h gives nonzero local vector pair vertices. The absence of quadratic mixing does not make these cubic terms vanish. The massless photon has no tree-level radial mass vertex, but charged-field loops can generate a photon-pair amplitude. Off-shell vector processes, supplied fermion channels and loop amplitudes are not computed in this release. In particular, closing on-shell WW and ZZ pairs does not prove a stable physical particle.

An exact cubic Ward identity. The radial fluctuation equation gives $h \ c_{GG} = E \ \psi_h + \psi_h \text{-double-prime} - (h \text{-double-prime} / h) \psi_h$. The normalized zero-energy global-angular continuum profile is h/h_0 . Multiplying by its square and integrating over the real line gives

$$\int c_{GG}(h/h_0)^2 d\rho = \frac{E}{h_0^2} \int h \psi_h d\rho + \frac{[h \psi_h' - h' \psi_h]_{-R}^R}{h_0^2}.$$

The finite endpoint term is retained. It is not set to zero merely because the radial mode is intended to be localized. This identity connects the scalar pair coupling to the radial Higgs component and its energy, and tests the mode equation through a quantity independent of the scalar Hessian expansion. The two sides agree with relative discrepancy about 2.3 times ten to the minus twelve in the physical-scale calculation. The nonzero moment demonstrates that the radial radiation-node tuning does not impose a generic cubic angular cancellation.

Normalization and interpretation of the reported overlaps. The source-mode derivative convention used by the preceding candidate solver is retained. If its full rho norm is N, the canonically normalized wall profile in physical z is $\sqrt{v/N} \psi(\rho)$, so integral over z of its squared magnitude is one. A bulk-pair vertex integrated against dimensionless continuum profiles then has units GeV to the one half. The zero-energy global-angular overlap is approximately 0.00366018549 in those units. The integrated W and Z mass-vertex coefficients, using constant diagnostic external profiles, are approximately 0.00196725353 and 0.00257069982. These quantities are vertices and Ward diagnostics, not on-shell decay amplitudes or widths. Their external profiles do not impose the energy conservation and distorted scattering wave functions needed for an actual two-particle decay calculation. A unit-scaling test verifies that masses scale with v and these overlaps with $\sqrt{v}$.

Numerical implementation and controls. The original finite-radius jointly tuned wall candidate supplies the quartics and an initial shape. A separate background solve and normalized real-mode boundary problem reconstruct its radial wave function at the selected quartics. The open channel's endpoint value is constrained, while its endpoint derivative is recorded independently. The mode is internally scaled by the source inverse width to avoid conditioning problems caused by the large source-normalization amplitude and heavy stiffness. This rescaling leaves the reported physical norm and overlaps unchanged. Domains of fourteen, eighteen and twenty-two source decay lengths retain all Ward boundary terms and integrated vertices. The profiles are sampled for the Higgs barrier and source range; all statements requiring continuum bounds retain their conditional status.

The gauge-input scan makes the kinematic dependence explicit. Reducing the declared g opens on-shell vector pairs without changing the tree spectator background, while the W integrated mass vertex scales exactly as g squared. This is a benchmark parameter scan, not evidence that measured gauge couplings vary. It distinguishes absent linear mixing from a kinematic closure that depends on the supplied vector masses. The static calculation includes no gauge thermal bath, vector determinant, fermion-loop matching or real-time production. Those belong to other calculations and must not be inferred from this channel analysis.

Delivery and reproduction. Twenty-four exact symbolic identities check the complete doublet Hessian, scalar pair vertices, Goldstone and longitudinal factorizations, Proca conjugation, kinetic derivative block, cubic Ward relation and neutral photon null direction. Ten focused tests cover those identities, the independent Ward integral, vector thresholds, nonzero cubic vertices, sampled barrier hypotheses, a longitudinal negative local well, declared-coupling dependence, normalization units and domain controls. Seventy-three related wall tests also pass. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_gauge_channels.py`, then `python scripts/plot_wall_gauge_channels.py` and `python scripts/render_wall_gauge_channels.py`. The source, receipts and illustrated monograph occupy separate wall files and preserve concurrent flavor and arithmetic work. There is no new Lean theorem, interval profile certificate or complete nonlinear lifetime in this release.

Relevant primary context is Takahiro Kubota, Gauge fields in the presence of the electroweak bubble wall, JHEP 03 (2026) 089, <https://arxiv.org/abs/2507.20134>. That work treats position-dependent electroweak masses, the coupled unphysical/gauge sector and the identification of physical polarization states. The present calculation derives its operators for a positive spectator Higgs barrier and proves the stated conditional quadratic-form bound; it does not claim to implement that paper's BRST quantization or bubble-wall friction calculation.